

Polynomials – 1

Roots, multiplicity and the derivative

Homework

8 сентября 2026 года

Theoretical minimum. Multiplicity of a root and the derivatives vanishing there; $\gcd(f, f')$ as the multiple part; Rolle's theorem for polynomials and the interlacing of the roots of f and f' ; Vieta's formulas; a non-negative real polynomial is a sum of two squares; the sign of a polynomial between consecutive roots.

1. (Putnam 1948) Let $f(x)$ be a cubic polynomial with roots x_1, x_2, x_3 and assume that $f(2x)$ is divisible by $f'(x)$. Compute the ratios of the roots.
2. (Putnam 1953) Assuming that the roots of $x^3 + px^2 + qx + r = 0$ are all real and positive, find the relation between p, q and r which is necessary and sufficient for the roots to be the cosines of the angles of a triangle.
3. (Putnam 1953) Show that the roots of $x^4 + ax^3 + bx^2 + cx + d = 0$, suitably numbered, satisfy $r_1/r_2 = r_3/r_4$ provided that $a^2d = c^2 \neq 0$.
4. (Putnam 1958) If $a_0, a_1, \dots, a_n$ are real numbers with $\sum_{k=0}^n \frac{a_k}{k+1} = 0$, show that the equation $a_0 + a_1x + \dots + a_nx^n = 0$ has at least one real root.
5. (Putnam 1978) The polynomials $Q_n(x)$ are defined by $Q_1(x) = 1 + x$, $Q_2(x) = 1 + 2x$ and, for $m \geq 1$, by $Q_{2m+1} = Q_{2m} + (m+1)xQ_{2m-1}$ and $Q_{2m+2} = Q_{2m+1} + (m+1)xQ_{2m}$. Let x_n be the largest real solution of $Q_n(x) = 0$. Prove that (x_n) is increasing and that $x_n \rightarrow 0$.

6. (Putnam 1981) Let $P(x)$ be a real polynomial and put $Q(x) = (x^2 + 1)P(x)P'(x) + x((P(x))^2 + (P'(x))^2)$. Given that $P(x) = 0$ has n distinct real roots exceeding 1, prove or disprove that $Q(x) = 0$ has at least $2n - 1$ distinct real roots.
7. (Putnam 1990) Does there exist an infinite sequence $a_0, a_1, a_2, \dots$ of non-zero real numbers such that for every n the polynomial $p_n(x) = a_0 + a_1x + \dots + a_nx^n$ has exactly n distinct real roots?
8. (Putnam 1941) Prove that the polynomial $(a - x)^6 - 3a(a - x)^5 + \frac{5}{2}a^2(a - x)^4 - \frac{1}{2}a^4(a - x)^2$ takes only negative values for $0 < x < a$.
9. (Putnam 2003) Suppose that a, b, c, A, B, C are real numbers with $a \neq 0$ and $A \neq 0$ such that $|ax^2 + bx + c| \leq |Ax^2 + Bx + C|$ for all real x . Show that $|b^2 - 4ac| \leq |B^2 - 4AC|$.
10. (Putnam 1946) If $a(x)$, $b(x)$, $c(x)$ and $d(x)$ are polynomials, show that $\int_1^x ac \cdot \int_1^x bd - \int_1^x ad \cdot \int_1^x bc$ is divisible by $(x - 1)^4$.